

# Suraj Kumar

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## EDUCATION

### Case Western Reserve University (CWRU)

Cleveland, OH

*Master of Science (M.S.), Computer Science*

*Aug 2026 - Dec 2027*

*Bachelor of Science in Engineering (B.S.E.), Biomedical Engineering*

*Aug 2023 - Dec 2027*

Relevant Coursework: Operating Systems (C), Intro to AI, Algorithms, Data Structures (Java), Discrete Math

## TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, JavaScript, HTML/CSS, SQL, Java, C

**Frameworks & Libraries:** Django, FastAPI, React.js, React Native, Next.js, PyTorch, TensorFlow, Scikit-learn, vLLM, HuggingFace, OpenCV, SQLAlchemy, Selenium

**Tools:** Git, AWS (ECS, VPC, RDS, S3), Docker, PostgreSQL, Linux, REST APIs, CI/CD, Gemini API, Claude Code

## SOFTWARE PROJECTS

AcoustiCare | Software Engineering Lead | *AWS, Docker, Django, React Native, Expo EAS* Nov 2025 – Present

- Led development of a voice-screening mobile app for ENT physicians that automates audio analysis of patient vocal health, **cutting diagnostic time from 2+ hours to seconds** per patient.
- Architected scalable AWS infrastructure (VPC, ECS, RDS PostgreSQL, S3, KMS) supporting **100+ users**, with layered security groups and TLS-terminated ALB to store sensitive clinical data at scale.
- Drafted the provisional patent white paper and established a clinical data collection pipeline with ENT physicians at UH Hospital to validate the system on real patient data.

Rival | Software Engineering Lead | *React Native, FastAPI, Gemini API, OAuth 2.0 (GitHub, Notion)* Jan 2026

- Built a GenAI-powered 1v1 mobile game using Gemini API with structured prompt engineering to score GitHub and Notion contributions via OAuth-integrated APIs, automatically determining who was more productive.
- Engineered the full-stack system (matchmaking engine, real-time progress tracking, and AI verdict pipeline) to handle concurrent multiplayer matches end to end.

## WORK/RESEARCH EXPERIENCE

### AI Security Graduate Researcher

May 2026 – Present

*Case Western Reserve University PEAT AI Lab*

*Remote/Cleveland, OH*

- Built an unsupervised log anomaly detection framework spanning authentication logs and intrusion-detection network flows, achieving **0.98+ AUROC** in real-time threat detection across **10M+ log events**.
- Engineered TF-IDF, Word2Vec, transformer and vLLM embedders (Qwen3) paired with four outlier detection models (GMM, KDE, One-Class SVM, DeepSVDD) to convert raw log streams into actionable anomaly alerts.
- Scaled preprocessing and embedding generation on Linux HPC clusters via chunked streaming I/O and parallel SLURM array jobs, **cutting full evaluation runtime by 4x**.

### Machine Learning Engineer Intern

Jun 2025 – Dec 2025

*Genentech*

*South San Francisco, CA*

- Reduced the Bioassay department's experimental workload by 30%** by training machine learning models and neural networks (PyTorch and Scikit-learn) to predict drug-potency experiment failures.
- Deployed ML inference endpoints to AWS EKS via GitLab CI/CD pipelines with **100%** automated test coverage.
- Optimized SQL queries across 30+ relational tables, **cutting data retrieval from 2 hours to 10 minutes**.
- Drove adoption of the analytics tool across **3 departments and 100+ employees** through technical demos to engineers, scientists, and stakeholders.

### Deep Learning Undergraduate Researcher

Sep 2024 – Nov 2025

*Case Western Reserve University INVent Lab*

*Remote/Cleveland, OH*

- Engineered a deep learning pipeline for glomeruli segmentation in pathology images that leverages a foundation model (MedSAM) to provide pseudo-labels for a UNet, **increasing baseline accuracy by 19%**.
- Automated extraction of 3,100 pathology slides from the KPMP database with Selenium, **cutting download time from 10 hours to 30 minutes**.
- Optimized DL algorithms on Linux HPC clusters by implementing cuDNN benchmarking and CUDA memory optimization strategies, **reducing training time by 3x**.